

NoPhobiar: Designing A VR Game to Prevent Childhood Dark Phobia with Children and Stakeholders

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Figure 1: NoPhobiar interactive storyboard.

ABSTRACT

Children's dread of the dark at night is a common occurrence as they develop, but if we disregard their worries without taking action, then they will develop a phobia of the dark, which will negatively impact their future. This study investigates the use of VR games in the prevention and treatment of dark phobia in youngsters. The majority of VR research, though, is not entirely appropriate for treating psychiatric disorders in young patients. We therefore invited children and their stakeholders to explore solutions to this issue in order to better grasp the issues and needs of treating children's phobia of darkness. In this paper, we develop a VR game for the prevention and treatment of dark phobia for children aged 4 to 6 years by engaging children and their stakeholders personally in our research to understand the design requirements and elements. Our research seeks to offer fresh design concepts for the VR treatment of mental illness in children between the ages of 4 and 6.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods.

KEYWORDS

Children's phobia of darkness, VR games, Participatory design

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1 INTRODUCTION

Phobia is very common in children between the ages of 4 and 6. It is mainly manifested in the fear of ghosts, insects, imaginary fear objects, especially when the child is in the dark environment, the fear of these objects becomes more intense. Although it is normal for children to experience short-term anxiety and fear in their development, if we do not intervene and allow children's fear to grow stronger, from fear and anxiety to phobia, it will directly affect children's future life. There are many different types of phobias that affect kids, but the focus of our research is on one in particular: dark phobia. Research have shown that specific phobias, which are a frequent mental health condition, have an 8% one-year prevalence in children and adolescents (including dark phobias) [1]. More than 73% of children aged 4 to 12 who participated in a study on Dutch children reported having nighttime phobias, with children in this age range reporting these anxieties the most frequently. It is worth noting that parents significantly underestimated the frequency of their children's night fears in this report [2]. This also causes children's fear of the dark to be easily ignored, which further leads to the generation of dark phobia.

The preferred treatment for phobias is exposure therapy [3, 4]. Exposure therapy involves exposing the feared object directly to the patient and confronting the fear until the pain is significantly reduced. With the increasing research in the field of VR treatment for psychological disorders, VR exposure therapy has been shown to be safe, practical [5], and very effective in treating specific phobias

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[6, 7]. Exposure therapy through games is widely used in children's psychotherapy. It not only has therapeutic effect on children with psychological and behavioral disorders, but also has a certain preventive effect on the general development problems of normal children, and has a high acceptance among children. For exposure treatment to help children overcome their fear of the dark, we used VR games in this study. Using VR games to cure dark phobia can help kids finish the therapy in a more acceptable way without putting them in danger because kids are typically more engaged in gaming technology and participate more actively.

In previous research, VR games have been studied for children's specific phobias, such as the use of VR games to alleviate children's autism [13, 25], children's spider phobia [20] and social phobia [19]. There are also many cases of games that specifically treat children's mental illness in real applications, such as EndeavorRX [8], EASE-Funhouse [9], SPARX [10]. At present, the visible shortcomings of VR games for the treatment of children's mental illness are in several aspects. For example, the overall visual style is realistic, and it is rarely designed according to children's aesthetic style. The interaction mode is complex. Previous studies have found that when children are allowed to play a more active role in developing children's platforms, they are more accepting [11]. The primary solution is to involve children and children's stakeholders participate in the design of VR games, as well as to draw inspiration from them while developing VR games for the treatment of mental illness and to assess each stage from their viewpoint.

In this paper, we will discuss VR game design considerations suitable for treating children's phobia of darkness through children aged 4 to 6 years, parents, experts and practitioners studying children, and VR professionals, and develop NoPhobia, a VR game based on free hand interaction to alleviate children's phobia of darkness (Figure 1). We first conducted an online questionnaire survey, in-depth brainstorming interviews with kids and stakeholders, and used previously published research knowledge to inform our design direction. Through the questionnaire and interview, we understand the circumstances in which children would have fear of the dark, and discuss it from the scene, fear object and interaction method. Then, based on the requirements of the above survey, we conducted a preliminary design development on the design points of NoPhobia, including VR environment design, fear object design and interaction design, and once again invited children, stakeholders and VR experts to evaluate and give suggestions on NoPhobia.

The key contributions of this work are as follows: (1) We investigate the demands and difficulties of VR games in reducing children's fear of the dark; and (2) Working with kids and other stakeholders, we created NoPhobia, a VR game to reduce kids' fears. (3) It has practical value and offers a fresh approach to using VR games to treat children's mental illnesses.

2 RELATED WORK

2.1 Childhood phobias

Although the use of VR to treat children's mental illness has been proved to be an effective and safe way to treat mental illness [5–7], most studies so far have focused on the training of skills learning for children with confirmed mental illness rather than the prevention or treatment of mental illness in children [12–15]. At present, only

a few VR games based on the prevention or treatment of children's anxiety phobia exist in the research. For example, Room VR [16] used VR games to study the treatment of children's bedtime anxiety and fear. Alice et al. [17] prevent and intervene the psychological and emotional health of children and adolescents through multi-scene VR games. Marieke et al. [18] used biofeedback virtual reality games (DEEP) to study the prevention and treatment of anxiety disorders in children. Willem-Paul et al. [19] used a virtual social VR scene to treat the fear of social phobia patients in daily social interaction. Wechsler et al. [20] developed a virtual reality serious game supporting eye tracking for school-age children to improve children's arachnophobia through gaze interaction. Up to now, there is no VR game specifically tailored for the prevention and treatment of such mental diseases for children aged 4 to 6 years old.

2.2 VR psychotherapy for children: freehand interaction and participatory design

Children in childhood are more active in body language, and appropriate interaction can bring better VR experience for children. VR research using hand interaction generally focuses on child education, skill learning, such as [12, 21–24]. In the current VR research, there are only a few studies related to the freehand interaction of VR in the treatment of children's psychological diseases. For example, Cai et al. [13] implemented immersive visualization and gear-based interaction through gesture learning (non-verbal) communication between autistic children and virtual dolphins to help children treat autism. Zhao et al. [14] promoted the communication and socialization of children with autism spectrum (ASD) through simple gesture collaboration. Battocchi et al. [25] used collaborative puzzle games (CPG) manipulated by direct finger touch to promote collaboration in children with ASD.

When we want to design a VR game that really contributes to children's mental illness, we should not just consider children as users, but involve children personally in the design process. This player-centric approach to co-designing games with the target audience is important because when we design games with children for children, Make them equal stakeholders rather than just users [26]. At the present stage of research, there is a lack of work to involve users and stakeholders in the design of treatments for mental illness. At present, the existing work on children's participatory design mainly includes, for example, Millen et al. [27] developed a Collaborative Virtual Environment (CVE) to promote the participatory design activities of ASC students. Vieira et al. [28] explored a child-centered design approach by engaging children actively in prototypical game design. Their results prove that when children are actively involved in prototyping, providing feedback and new ideas, serious games can be made more engaging and easier for other children. Cognality VR [29] designed an application to help children with ASC communicate with others by collaborating with stakeholders of children with autism. vrSensory[30] designed inclusive VR games by working with neurodiverse children. In these works, we hope to draw on the experience of participatory design and involve children and stakeholders in the design of our VR games.

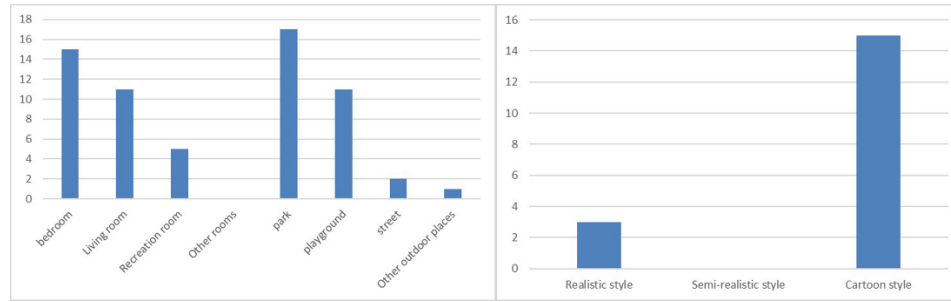


Figure 2: The left figure shows the results for question 3 and the right figure shows the results for question 4.

3 DESIGN EXPLORATION

3.1 Research objectives

To ascertain the design needs of virtual fear species, key scenes, virtual object interaction dynamics, and the placement of fear objects in the scene in VR games, we first performed an online questionnaire on children’s phobia. In order to comprehend the design requirements and components of children’s phobia from various angles and perspectives, we conducted interviews with research specialists or teachers who were closely associated to children as well as children aged 4-6 (who were aided by their parents). Our study’s specific goals are to: (1) compare different demographic views to assess the balance between children’s realistic demands for phobia relief and design aspects in VR games; and (2) brainstorm solutions with targeted users (kids aged 4-6 years) and stakeholders.

3.2 Interview design

We recruited a total of 18 participants to take part in the preliminary design analysis in order to fine-tune the specific design components of the VR game. There were six youngsters, ages 4-6, along with six parents and six teachers. We classified the locations visited by kids into two categories in the interview questions: indoor locations and outdoor locations at home; In terms of game scene design, we gave interview participants a variety of images to choose from when choosing an appropriate VR game scene design style for kids. The subjects of this questionnaire and interview have filled in informed consent, and their privacy was well protected. The following topics are primarily covered in the interview’s questions: (1) What do you think scares children (ask non-children)/What do you think scares you (ask children)? (2) What are the main methods you use to make yourself less afraid of these items (ask children)? (3) Which of the following places do you think children aged 4-6 often go to (ask non-children)/which of the following places do you often go (ask children)? Family indoor places: a. Bedroom b. Living room c. Recreation room d. Other rooms; Outdoor places: a. Park b. Playground c. Street d. Other outdoor places; (4) Please select the following picture style that you think children aged 4-6 like (ask non-children)/Please select your favorite picture style (ask children).

3.3 Interview results

According to the interview findings, the most common location for kids in indoor scenes is a bedroom, and the most popular location for kids in outdoor scenes is a park (Figure 2 left). Most participants

selected a scenario that was more cartoon-like in terms of the picture’s general style design (Figure 2 right), and we will examine the causes in more detail in the discussion that follows.

Virtual Fear Objects: When it comes to fear objects, some parents of young children have claimed that their children occasionally fear things that are actually not frightening or even unusual, such as curtain fabric, hanging clothes, or their own toys. It’s possible, however, that the silhouette of these objects in a dark environment will occasionally cause children to feel strange and frightened. One youngster revealed that he occasionally thought of intangible items as ghosts or monsters, that he was particularly terrified of floating or horned objects, and that he even thought of dark sources of light as terrifying eyes. Another child claimed that despite not seeing anything, he was afraid when he heard a peculiar noise. For fear of spiders or other real objects, some children said, “These spiders look scary”. This inspired us to focus on the visual and auditory changes of the fear object.

Game Scene Layout: According to the majority of people’s responses, we decided to build the game scenario with the bedroom serving as the indoor scene and the park serving as the outdoor scene. We did in-depth user interviews in order to construct these two scenarios in further detail. According to several parents, their kids were more terrified in dimly lit situations than in bright ones. “In the bedroom, when I lay alone in bed, if my mother is not around, I feel scary things come out from under the bed, behind the window or in the cupboard, and sometimes I feel like something is watching me in the corner of the room,” one child said in more detail. Another child added to the details of the park fear: “When I go to the park in the evening, I think there is something in the forest in the park, I feel a little afraid, my mother also said not to go to such a place, there will be a monster to seize me”. Parents claimed that they frequently warn their children about the dangers of specific dark and occluded regions in order to keep their children from running too far away from them or disappearing out of sight. Also, this vocal warning makes the child’s psychological anxiety worse. The interviews mentioned above provided us with plenty of ideas for where to put the fear-inducing things in the scenario and how to help them to feel less afraid by speaking encouraging words aloud to them.

Game Scene Style: We offered the options of a fully realistic photo scene, a somewhat realistic modeled scene, and a cartoon scenario; the majority of users selected the cartoon scene. According to a specialist in children’s education, cartoons in abstract styles are

more appealing to kids, and when compared to real scenes, cartoon scenes that are succinct can help kids grasp the information in the scene more readily. All of the kids pointed to the cartoon image and said, "I like this one better." As a result, we made the decision to create the entire VR environment in a more cartoonish and juvenile manner.

Interaction: We interviewed users about how they hoped children would interact with fear objects to ease their fears. Some parents and children's education experts said that if children could interact with fear objects more, such as saying hello, smiling and chatting, they might be able to ease their fears. One child specifically said, "I want to make things that are afraid become things that are not afraid, such as monsters become superman". As a result, we made the decision to expand our design to include more interactive encounters with virtual fear objects.

4 EVALUATION: USER STUDY

Inspired by the results of the questionnaire and interview, we designed NoPhobia, a VR game to alleviate children's phobias in a dual scene of bedroom and park, and added an interactive experience of interacting with virtual objects through gestures to the game. To assess the role of NoPhobia in alleviating phobias in children and to understand how NoPhobia is rated, we conducted a user study. Since the local area is still affected by COVID-19, it is impossible to conduct face-to-face user evaluation and experience, we show the whole interaction process and design details to the evaluators in the form of video pictures online, and conduct in-depth interviews with them. The above user study were conducted with the consent of the users and filled out the informed consent form, and the user privacy was well protected.

4.1 Design approach

NoPhobia is a VR game developed and built through the Unity2019 engine. The VR head-mounted display is used as the visual presentation device of the VR game, Leap motion is used as the hand interaction recognition device, and C4D modeling is used. By using VR headsets, kids can engage with the virtual environment with their bare hands. In a dark setting, the child's left hand serves as a spotlight to find the fright object using sound and visual as guides. In order to reveal the treatment child's fear, the fear object changes as a result of the child's interactions with it and provides feedback. Scene 1 establishes the setting for our tale: at night, the child sleeps by himself in the bedroom. The young boy uses a flashlight to look around the room as he fights his fear of the room's black shadow. Scene 2: The child is waiting for the parents alone at the park after the parents leave for the evening. The child shines a light to illuminate the dark shadow in the pitch-black area where there appears to be a peculiar sound coming from in the park.

Scene Design: Scenes 1 and 2 are set at night and evening respectively (as shown in Figure 3), so the overall environment is relatively dim. In Scene 1 bedroom, the user is in the center of the bedroom, and the virtual fear objects are placed around the scene: in a half-open cabinet, under the bed, behind the curtains, and in a dark and cluttered corner. In Scene 2, the user is still in the center of the park, but since the park area is larger than the indoor bedroom area, the virtual fear objects are concentrated directly in the front



Figure 3: The park scene is shown on the right and the bedroom scene is shown on the left.



Figure 4: Virtual fear object design.

of the screen: the dark bushes and the night sky. The two scenes are arranged with four fear objects respectively, and the whole scene is designed in a relatively cartoon style.

Virtual Fear Object Design: We summarized the fear objects in the preliminary design elements interview and mainly designed the following kinds. The virtual fear objects in the bedroom are (as shown in Figure 4): rabbits, dolls, floating ghosts, spiders; The virtual fear objects in the park are (shown below in Figure 4): bats, big-eyed owls, cows, and black cats. Virtual fear objects are divided into two categories: one is not scary itself but the silhouette will cause children to be afraid, such as horned cattle and rabbits; Another category is the fear objects that are scary in themselves such as dolls, floating ghosts, spiders, bats, owls, and black cats. After children enter the scene, the virtual fear object first appears in the environment in the form of silhouette, and sends out the voice "guess where I am" to guide children to search for the fear object.

Interactive Design: The left hand of the child was used as a flashlight, and the movement of the left hand controlled the activity of the light (Figure 1). When children hit the flashlight on the virtual fear object, the virtual fear object will show its original shape after 3 seconds. If it is a scary virtual object (such as a ghost), the light will be hit on the virtual fear object for 8 seconds, and the fear object will change from a scary form to a lovely form and make a sound, "In fact, I am not so scary, right?". The child interacts with the virtual fear object through his right hand. When the child waves at the virtual object, the virtual object waves back to the child and makes a speech "Hello".

Following is the specific interaction process: 1. After wearing the VR helmet, kids select either the bedroom or park scene to explore. Children enter the virtual reality setting, move their heads to explore the surroundings, and then use a left-handed torch to illuminate the area. 2. In response to the sound "Guess where I am,"

children search their surroundings for the silhouettes of frightful items. 3. When the dread object is illuminated by a child's flashlight, it emits the phrase "Don't be scared," returns to its original form after three seconds, then, after eight seconds, transforms into a charming image and says, "In fact, I'm not so horrible." 4. Children wave their right hand to greet the fear object, and the fear object waves back "Hello".

4.2 Research questions and measures

The purpose of this study was to determine how effectively NoPhobia, a game that helps youngsters overcome their phobias, performed overall and how different groups assessed NoPhobia. Six children between the ages of four and six, six parents, six researchers or instructors with knowledge of children's research, two VR experts—a total of 20 interviewees—make up our interview group. We questioned the interviewees on NoPhobia's usability, overall experience, and detailed experience, among other factors. The following were the primary interview questions: (1) After learning about NoPhobia, from your perspective, do you think this game can help children with phobias, and how does it perform in terms of usability? (2) Please give your overall experience and evaluation of NoPhobia, and point out the pros and cons and why; (3) Please comment and suggest scenarios, objects and interactions for NoPhobia.

4.3 Result

We interviewed the evaluation of our VR game from different groups, and the suggestions of different groups gave us great inspiration. They all represented the views of this group on the VR game to treat children's dark phobia to a certain extent.

Children aged 4-6 years: With the help of parents, we completed interviews with six children, all of whom had experienced fear of the dark. After we showed our VR game to the children, they showed clear interest. All the children expressed their love for the overall design of the game style. For example, one child mentioned that "This game is very similar to the cartoons I usually watch", and another child pointed out that the interactive changes of virtual fear objects can alleviate his fear. "These are all things I usually fear, but I seem to be less afraid after they are transformed into them".

Parents of children: Most interviewers spoke highly of NoPhobia, especially parents. All the parents in the interviewer agreed that NoPhobia can help most children prevent and alleviate dark phobia after learning about the use of our VR games. The father of one of the children pointed out, "Good idea! Because there is a step-by-step process, let the child from fear to not fear, gradually accept fear, eliminate fear. I'd love to let my kids play it!". Some parents also put forward constructive suggestions. For example, one mother said, "I think the fear level of these fear objects is not so scary, and objects with different fear levels should be added. There can be more ferocious animals in the scene, or animals can suddenly appear, such as snakes or dogs in the park, to make the objects more vivid". There is also a parent who hopes that our design can be replaced with a lighter device for children to experience, especially if they can directly start the game without special equipment.

Experts or teachers with research experience in children: Most of the respondents recognized the positive effects of NoPhobia

and believed that it could treat and prevent children's dark phobia to some extent. Some experts and teachers think that the biggest advantage of this VR game is that the overall design style is very suitable for children aged 4-6. Whether it is the scene design of bedrooms and parks or the image design of virtual fear objects, it is full of childlike innocence and fun. They think this is the key for children to accept and experience this game. However, some respondents also questioned whether the image design of some fear objects is too cartoon, and will it lose the effect of fear? "For example, children are not afraid of the cartoon spider in this game, but they are still afraid of the real spider, because the real spider is scarier". In terms of interaction, an expert mentioned that "The interaction of this game is very good and interesting. Using" waving to say hello "as a gesture is very humane, and it is also in line with the way children interact with other animals, but I hope that the waving feedback animation of animals can be made more lively".

VR experts: We also invited VR experts to evaluate and advise on our VR game. Some of the experts made recommendations for the two VR scenarios: although the fear objects in the bedroom are more dispersed due to the spatial difference between the bedroom and the park, the fear objects in the park are relatively concentrated. However, this will cause children's viewing styles to change in the two scenes. In the bedroom scene, children need to constantly rotate their heads to explore, while they only need to focus on one place in the park, which may lead to children switching from one scene to another, using the viewing style of the previous scene directly in the next scene by default, affecting subsequent experience and interaction. Another expert also asked us for suggestions from other groups. He pointed out that the way of increasing popular science voice interaction mentioned by preschool teachers is very interesting, which can appropriately increase children's interactive gesture rich experience.

5 LIMITATION & FUTURE WORK

Our work currently has several regrets and limitations. First of all, due to the impact of COVID-19, the number of children and stakeholders we invited was relatively small, and most of them could only communicate online. Although everyone showed a high level of participation, we hope to have the opportunity to discuss with you offline in future work. Secondly, we failed to fully develop NoPhobia as a VR game, but only showed the key steps and processes to children and stakeholders. We were unable to examine the actual impact of NoPhobia on the management of children's dark fear due to the epidemic. In the future work, we will continue to invite children and stakeholders to participate in our next stage of design, NoPhobia is fully developed and can be deployed on various platforms without being limited by most devices.

6 CONCLUSIONS

In this research work, we invited children and stakeholders to participate in the design of NoPhobia, a VR game for treating children's dark phobia. The goal is to discuss the design elements of this VR game from the perspectives and needs of various groups in order to provide a truly suitable design for kids. We discovered that engaging children and stakeholders in the design process rather than just using them as research subjects generated many fresh design

concepts and improved child acceptance of the VR game, which improved its therapeutic effects for dark phobia. Our research aims to offer fresh design concepts for VR work treating psychological disorders and new approaches to treating children's phobia of the dark.

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REFERENCES

- [1] Soo-Jin Kim, Bung-Nyun Kim, Soo-Churl Cho, Jae-Won Kim, Min-Sup Shin, Hee-Jung Yoo, and Hyo Won Kim. 2010. The prevalence of specific phobia and associated co-morbid features in children and adolescents. *Journal of Anxiety Disorders* 24, 6 (Aug 2010), 629–634.
- [2] Muris P, Merckelbach H, Ollendick TH, King NJ, Bogie N. Children's nighttime fears: parent-child ratings of frequency, content, origins, coping behaviors and severity. *Behav Res Ther*. 2001 Jan;39(1):13-28. doi: 10.1016/s0005-7967(99)00155-2. PMID: 11125721.
- [3] Dianne L. Chambless, William C. S. Suzanne Bennett Johnson, Kenneth S. Pope, Mary Baker, Benjamin Johnson, Sheila R, Stanley Sue, Larry Beutler, David A. Williams, and Susan McCurry. 1998. Update on empirically validated therapies. II. *Clinical Psychologist* (1998), 3–16.
- [4] Kate B. Wolitzky-Taylor, Jonathan D. Horowitz, Mark B. Powers, and Michael J. Telch. 2008. Psychological approaches in the treatment of specific phobias: A meta-analysis. *Clinical Psychology Review* 28, 6 (Jul 2008), 1021–1037. <https://doi.org/10.1016/j.cpr.2008.02.007>
- [5] Javier Fernández-Álvarez, Alexander Rozenal, Per Carlbring, Desirée Colombo, Giuseppe Riva, Page Anderson, Rosa Baños, Amanda Draheim, Stéphane Bouchard, Juana Bretón-López, Georgina Cárdenas, Joann Difede, Paul Emmelkamp, Azucena García-Palacios, Verónica Botella, Hunter Hoffman, Isabel Kampmann, Ramona Moldovan, Andreas Mühlberger, and Cristina Botella. 2019. Deterioration rates in Virtual Reality Therapy: An individual patient data level meta-analysis. *Journal of Anxiety Disorders* 61 (01 2019), 3–17. <https://doi.org/10.1016/j.janxdis.2018.06.005>
- [6] Emily Carl, Aliza T. Stein, Andrew Levihn-Coon, Jamie R. Pogue, Barbara Rothbaum, Paul Emmelkamp, Gordon J.G. Asmundson, Per Carlbring, and Mark B. Powers. 2019. Virtual reality exposure therapy for anxiety and related disorders: A meta-analysis of randomized controlled trials. *Journal of Anxiety Disorders* 61 (Jan 2019), 27–36. <https://doi.org/10.1016/j.janxdis.2018.08.003>
- [7] Theresa F. Wechsler, Franziska Kümpers, and Andreas Mühlberger. 2019. Inferiority or Even Superiority of Virtual Reality Exposure Therapy in Phobias? - A Systematic Review and Quantitative Meta-Analysis on Randomized Controlled Trials Specifically Comparing the Efficacy of Virtual Reality Exposure to Gold Standard in vivo Exposure in Agoraphobia, Specific Phobia, and Social Phobia. *Frontiers in Psychology* 10 (Sep 2019), 1758. <https://doi.org/10.3389/fpsyg.2019.01758>
- [8] EndeavorRx. 2023. EndeavorRx: A Prescription Treatment for ADHD. Retrieved February 27, 2023 from <https://www.endeavorrx.com/>
- [9] EASE Funhouse Demo. 2023. EASE Funhouse Demo Download (EASEFunhouse.exe). Retrieved February 27, 2023 from <https://ease-funhouse-demo.software.informer.com/>
- [10] Sparx. 2023. Home - Sparx. Retrieved February 27, 2023 from <https://www.sparx.org.nz/home>
- [11] My-Linh Ho, Dhanush kumar Ratakonda, and Jerry Alan Fails. 2022. KidConnect VR: Technology to Stay Connected. In *Interaction Design and Children (IDC '22)*. Association for Computing Machinery, New York, NY, USA, 645–648. <https://doi.org/10.1145/3501712.3536391>
- [12] A. Adjorlu, E. R. Høeg, L. Mangano and S. Serafin, "Daily Living Skills Training in Virtual Reality to Help Children with Autism Spectrum Disorder in a Real Shopping Scenario," 2017 IEEE International Symposium on Mixed and Augmented Reality (ISMAR-Adjunct), 2017, pp. 294-302, doi: 10.1109/ISMAR-Adjunct.2017.93.
- [13] Y. Cai, N. K. H. Chia, D. Thalmann, N. K. N. Kee, J. Zheng and N. M. Thalmann, "Design and Development of a Virtual Dolphinarium for Children With Autism," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 21, no. 2, pp. 208-217, March 2013, doi: 10.1109/TNSRE.2013.2240700.
- [14] H. Zhao, A. R. Swanson, A. S. Weitlauf, Z. E. Warren and N. S. Sarkar, "Hand-in-Hand: A Communication-Enhancement Collaborative Virtual Reality System for Promoting Social Interaction in Children With Autism Spectrum Disorders," in *IEEE Transactions on Human-Machine Systems*, vol. 48, no. 2, pp. 136-148, April 2018, doi: 10.1109/THMS.2018.2791562.
- [15] U. Lahiri, Z. Warren and N. S. Sarkar, "Design of a Gaze-Sensitive Virtual Social Interactive System for Children With Autism," in *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 19, no. 4, pp. 443-452, Aug. 2011, doi: 10.1109/TNSRE.2011.2153874.
- [16] Vy Dang Ha Thanh, Ondris Pui, and Martin Constable. 2017. Room VR: a VR therapy game for children who fear the dark. In *SIGGRAPH Asia 2017 Posters (SA '17)*. Association for Computing Machinery, New York, NY, USA, Article 52, 1–2. <https://doi.org/10.1145/3145690.3145734>
- [17] Alice J Lin, Charles B Chen, Fuhua (FRANK) Cheng. 2022. Improving the Emotional Health of Juveniles with VR Game. In *2022 the 14th International Conference on Computer Modeling and Simulation (ICCMS 2022)*, June 24-26, 2022, Chongqing, China. ACM, New York, NY, USA, 11 Pages. <https://doi.org/10.1145/3547578.3547608>
- [18] Marieke van Rooij, Adam Lobel, Owen Harris, Niki Smit, and Isabela Granic. 2016. DEEP: A Biofeedback Virtual Reality Game for Children At-risk for Anxiety. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems (CHI EA '16)*. Association for Computing Machinery, New York, NY, USA, 1989–1997. <https://doi.org/10.1145/2851581.2892452>
- [19] Willem-Paul Brinkman, Dwi Hartanto, Ni Kang, Daniel de Vliegheer, Isabel L. Kampmann, Nexhmedin Morina, Paul G.M. Emmelkamp, and Mark Neerincx. 2012. A virtual reality dialogue system for the treatment of social phobia. In *CHI '12 Extended Abstracts on Human Factors in Computing Systems (CHI EA '12)*. Association for Computing Machinery, New York, NY, USA, 1099–1102. <https://doi.org/10.1145/2212776.2212395>
- [20] Theresa F. Wechsler, Martin Brockelmann, Konstantin Kulik, Felicitas M. Kopf, Martin Kocur, Michael Lankes, Andreas Mühlberger, and Christian Wolff. 2021. SpEYEders: Adults' and children's affective responses during immersive playful gaze interactions transforming virtual spiders. In *Extended Abstracts of the 2021 Annual Symposium on Computer-Human Interaction in Play (CHI PLAY '21)*, October 18–21, 2021, Virtual Event, Austria. ACM, New York, NY, USA 6 Pages. <https://doi.org/10.1145/3450337.3483463>
- [21] Krzysztof Pietroszek and Liudmila Tahai. 2019. Scalebridge VR: Immersive Proportional Reasoning Game for Children with Brain-Computer Interface for Difficulty Scaling. In *25th ACM Symposium on Virtual Reality Software and Technology (VRST '19)*, November 12–15, 2019, Parramatta, NSW, Australia. ACM, New York, NY, USA 2 Pages. <https://doi.org/10.1145/3359996.3364734>
- [22] Gökçe Elif Baykal, Ali Leylekoğlu, Can Bora Sezer, and İslim Oygür. 2022. Studying Children's Manipulative Gestures in Spatial Puzzle Play with VR Hand Tracking: Analysis of Goal-directed Actions. In *Symposium on Spatial User Interaction (SUI '22)*, December 1–2, 2022, Online, CA, USA. ACM, New York, NY, USA 2 Pages. <https://doi.org/10.1145/3565970.3567707>
- [23] S. Shimada, K. Funahashi, K. Ito, Y. Tanase and K. Iwazaki, "VR hand hygiene training system that visualizes germs to be washed and removed," 2017 IEEE 6th Global Conference on Consumer Electronics (GCCE), 2017, pp. 1-4, doi: 10.1109/GCCE.2017.8229333.
- [24] A. Karkar, T. Salahuddin, N. Almaadeed, J. M. Aljaam and O. Halabi, "A Virtual Reality Nutrition Awareness Learning System for Children," 2018 IEEE Conference on e-Learning, e-Management and e-Services (IC3e), 2018, pp. 97-102, doi: 10.1109/IC3e.2018.8632634.
- [25] A. Battocchi et al., "Collaborative puzzle game: A tabletop interactive game for fostering collaboration in children with Autism Spectrum Disorders (ASD)", *Proc. ACM Int. Conf. Interactive Tabletops Surf.*, pp. 197-204, 2009.
- [26] John Edison Muñoz Cardona, Shruti Chandra, Adriana Rios Rincon, Luke Jai Wood, and Kerstin Dautenhahn. 2021. Designing Games for and with Children. Co-design Methodologies for playful activities using AR/VR and Social Agents. In *Interaction Design and Children (IDC '21)*, June 24–30, 2021, Athens, Greece. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3459990.3460517>
- [27] Millen, Laura, Sue Valerie Gray Cobb, Harshada Patel and Tony Glover. "Collaborative virtual environment for conducting design sessions with students with autism spectrum conditions." (2012).
- [28] Sandrina Vieira, Isabel Machado Alexandre and Elsa Cardoso. 2022. The Child in Charge: The Case of Child-participatory Design of an Environmental Sustainability Serious Game. In *Interaction Design and Children (IDC '22)*, June 27-30, 2022, Braga, Portugal. ACM, New York, NY, USA, 9 Pages. <https://doi.org/10.1145/3501712.3535314>
- [29] Louanne E. Boyd, Espen Garner, Ian Kim and Gianna Valencia. 2022. Cognality VR: Exploring a Mobile VR App with Multiple Stakeholders to Reduce Meltdowns in Autistic Children. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 11 Pages. <https://doi.org/10.1145/3491101.3519742>
- [30] Ben Wasserman, Derek Prate, Bryce Purnell, Alex Muse, Kaitlyn Abdo, Kendra Day, and LouAnne Boyd. 2019. VrSensory: Designing Inclusive Virtual Games with Neurodiverse Children. In *Extended Abstracts of the Annual Symposium on Computer-Human Interaction in Play Companion Extended Abstracts (CHI PLAY '19 Extended Abstracts)*. Association for Computing Machinery, New York, NY, USA, 755–761. <https://doi.org/10.1145/3341215.3356277>